

PROVIDENCE GREEN, RI, USA

WMO: 725070

Lat: 41.723N

Lon: 71.433W

Elev: 17

StdP: 101.12

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14765

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-10.7	-23.4	0.5	-11.2	-20.9	0.6	-8.7	12.7	3.9	11.5	2.6	4.5	310	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	32.2	23.0	30.5	22.3	28.9	21.4	24.8	29.7	23.9	27.9	23.1	26.6	5.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.1	26.9	22.6	17.4	26.1	22.0	16.7	25.3	75.0	29.9	71.6	27.9	68.5	26.7	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.2	8.3	-16.4	35.3	3.0	1.8	-18.6	36.6	-20.3	37.6	-22.0	38.6	-24.1	39.9		
(5)			-17.4	26.2	2.8	0.9	-19.4	26.8	-21.0	27.3	-22.6	27.8	-24.6	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-1.1	0.0	3.8	9.6	14.9	20.0	23.5	22.7	18.8	12.6	7.0	2.1	(6)
(7)		DBStd	9.49	5.57	5.03	4.70	4.00	3.91	3.63	2.85	2.85	3.41	4.07	4.54	4.84	(7)
(8)		HDD10.0	1264	346	282	200	53	3	0	0	0	19	113	249		(8)
(9)		HDD18.3	3043	603	514	451	262	123	24	1	2	36	183	341	503	(9)
(10)		CDD10.0	1707	1	1	7	42	154	301	419	394	263	99	22	4	(10)
(11)		CDD18.3	443	0	0	0	1	16	74	162	137	49	5	0	0	(11)
(12)		CDH23.3	3261	0	0	3	22	151	538	1273	987	264	24	0	0	(12)
(13)		CDH26.7	979	0	0	0	7	43	158	422	293	53	2	0	0	(13)
(14)	Wind	WSAvg	4.0	4.2	4.3	4.6	4.5	4.1	3.8	3.6	3.4	3.5	3.8	3.9	4.0	(14)
(15)	Precipitation	PrecAvg	1154	100	88	109	102	91	83	74	94	98	96	108	110	(15)
(16)		PrecMax	1715	296	183	224	324	213	281	205	256	201	391	280	273	(16)
(17)		PrecMin	646	13	10	14	38	18	4	10	18	23	10	10	26	(17)
(18)		PrecStd	193	55	37	52	52	45	57	41	55	49	70	53	58	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.6	15.5	21.0	26.7	31.2	33.2	34.8	33.9	30.9	26.2	20.5	17.5	(19)
(20)			MCWB	13.6	11.8	13.0	16.6	20.4	23.4	24.3	24.3	22.7	19.1	15.8	15.6	(20)
(21)		2%	DB	12.6	12.1	16.2	21.5	27.3	30.5	32.5	31.4	28.0	23.1	18.0	14.3	(21)
(22)			MCWB	10.4	8.7	10.3	13.6	18.6	22.0	23.3	23.0	20.8	18.6	14.8	12.0	(22)
(23)		5%	DB	9.6	9.6	13.1	18.6	24.4	28.4	30.6	29.6	26.2	21.2	16.3	12.1	(23)
(24)			MCWB	7.4	6.7	8.3	12.0	17.0	20.6	22.6	21.9	20.1	17.2	13.3	10.1	(24)
(25)		10%	DB	6.8	7.3	10.7	16.3	21.9	26.4	28.9	28.2	24.6	19.4	14.4	9.8	(25)
(26)			MCWB	4.6	4.4	6.9	10.6	15.6	19.8	21.8	21.4	19.5	15.8	11.6	7.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.3	13.2	14.8	17.8	21.9	24.8	25.9	25.7	24.1	21.8	17.8	15.5	(27)
(28)			MCD B	15.2	14.6	19.3	22.6	29.2	30.7	32.0	31.9	27.3	23.9	19.3	16.7	(28)
(29)		2%	WB	10.9	9.9	11.8	15.4	20.0	23.4	25.0	24.6	22.8	19.7	15.8	12.7	(29)
(30)			MCD B	11.9	11.5	14.7	19.2	25.4	28.4	30.3	29.0	25.6	21.8	17.2	13.9	(30)
(31)		5%	WB	7.7	7.1	9.7	13.6	18.4	22.2	24.0	23.8	21.9	18.1	14.0	10.4	(31)
(32)			MCD B	9.3	9.0	12.3	16.9	22.9	26.3	28.2	27.3	24.6	20.4	15.6	11.7	(32)
(33)		10%	WB	4.9	4.7	7.4	11.9	16.8	21.1	23.2	23.0	21.0	16.5	12.1	7.8	(33)
(34)			MCD B	6.6	6.8	10.1	14.8	20.4	24.6	27.0	26.0	23.5	19.0	14.0	9.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	8.6	8.9	9.8	9.9	9.5	9.3	9.3	9.5	8.8	7.9	(35)	
(36)			MCDBR	11.3	11.2	12.4	13.5	13.8	12.0	11.0	10.8	10.7	11.1	10.3	10.8	(36)
(37)			MCWBR	10.2	9.1	8.2	7.3	6.9	5.9	4.5	4.5	5.6	7.0	8.2	9.7	(37)
(38)		5% WB	MCD BR	11.1	10.9	11.5	11.6	12.5	10.5	9.5	9.0	8.7	9.5	9.3	10.3	(38)
(39)			MCWBR	10.6	10.0	9.1	7.8	7.0	5.7	4.3	4.5	5.5	7.4	8.7	10.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.308	0.332	0.372	0.414	0.460	0.483	0.441	0.373	0.338	0.311	0.298	(40)	
(41)		taud	2.499	2.480	2.463	2.381	2.296	2.206	2.150	2.255	2.420	2.530	2.564	2.560	(41)	
(42)		Ebn at Noon	858	905	918	898	863	821	797	821	861	858	839	831	(42)	
(43)		Edh at Noon	73	87	99	116	129	142	149	130	103	82	69	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.87	2.74	3.75	4.82	5.33	5.82	6.01	5.40	4.39	2.95	2.02	1.56	(44)	
(45)		RadStd	0.14	0.26	0.27	0.52	0.54	0.58	0.33	0.28	0.32	0.27	0.15	0.15	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86	+51
(47) Regional (70 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+46

Nomenclature: See separate page